Time Table for Thesis Project

Week	Main task	Output
Week 1	Study related literature and define the structure of the thesis	Clear thesis outline and reading notes
Week 2	Write **Introduction and Related Literature**	First draft of Introduction + Literature Review
Week 3	Improve the literature section, especially weak instruments and confidence regions	Stronger literature review with focus on weak identification and robust inference
Week 4	Write the theoretical framework: IVQR, structural quantile effects, identification assumptions	Draft of Methodology section
Week 5	Write the simulation design: DGPs, parameters, true effects, estimators, and evaluation metrics	Complete Simulation Design section
Weeks 6–8	Write the full project code and run simulations	Working code, Monte Carlo results, tables, graphs
Week 9	Work with empirical data and apply the estimators	Empirical application section + preliminary results
Week 10	Write the Simulation Results section	Draft of Results section
Week 11	Write conclusion, limitations, robustness checks	Full first draft of thesis
Week 12	Final editing, formatting, references, and preparation for professor's questions	Final version + defence notes

Week 1 — Literature Preparation

Study the main related literature:

- IVQR model and structural quantile treatment effects.
- DML and Neyman orthogonality.
- DML-IVQR with high-dimensional controls.
- Weak instruments and weak-identification robust inference.

Week 2 — Introduction and Related Literature

Write:

- motivation of the thesis;
- research questions;
- why IVQR is useful;
- why high-dimensional controls create a problem;
- why DML/post-selection methods are relevant.

Week 3 — Weak Instruments and Confidence Regions

Add a focused discussion on:

- weak instruments in IVQR;
- why point estimates may be unreliable under weak identification;
- why confidence intervals may be misleading;
- why weak-identification robust confidence regions are important.

Week 4 — Methodology

Write the methodology section:

- standard IVQR;
- full-control IVQR;
- post-selection IVQR;
- DML-IVQR;
- role of high-dimensional controls;
- role of orthogonal moments.

DML is based on sample splitting/cross-fitting and Neyman orthogonality, which makes inference less sensitive to first-stage nuisance estimation.

Week 5 — Simulation Design

Fix the complete simulation design:

- DGP1: baseline design;
- DGP2: higher sparsity / more relevant controls;
- DGP3: weak instruments;
- sample sizes;
- number of controls;
- number of Monte Carlo repetitions;
- true structural quantile treatment effect;
- metrics: median bias, MAE, RMSE, non-convergence rate, possibly coverage.

Weeks 6–8 — Coding and Running Simulations

Use three full weeks for:

- writing clean code;
- testing each estimator separately;
- debugging convergence problems;
- running small simulations first;
- running full Monte Carlo simulations;
- saving tables and graphs automatically;
- checking whether results are stable.

Week 9 — Empirical Data

Add an empirical application:

- describe the dataset;
- define outcome, treatment, instruments, and controls;
- apply the estimators;
- compare empirical results with simulation findings;
- discuss whether the empirical results show signs of weak identification.

Week 10 — Simulation Results

Write the results section:

- compare estimators across DGPs;
- explain bias, MAE, RMSE, and convergence;
- discuss performance under weak instruments;
- interpret confidence regions;
- explain why DML-IVQR should perform better with many controls.

Week 11 — Conclusion and Limitations

Write:

- main findings;
- what DML-IVQR improves;
- where it still fails;
- limitations: computational cost, convergence, weak instruments, finite-sample instability;
- possible extensions.

Week 12 — Final Checking

Finalize:

- formatting;
- references;

- tables;
- appendix;
- code documentation.